

PATHSEEKER CAREER PASSPORT

# Junior Data Analyst 90-Day Roadmap

A project-first plan for building spreadsheet, SQL, visualization, statistics, and communication evidence in twelve focused weeks.

## BEST FOR

Students, graduates, and career changers targeting junior data analyst or business intelligence roles.

## FORMAT

9-page roadmap - write directly on a printed copy or duplicate the prompts into your notes app.

**Discover what fits you best - then test it with evidence.**

## 01 - WEEK 0

# Set your baseline

Start with a target job sample and a diagnostic. The goal is not to learn every tool; it is to build evidence for a realistic entry role.

- [ ] Collect 10 relevant job descriptions.
- [ ] Count repeated skills and tools.
- [ ] Choose one spreadsheet tool, one SQL environment, and one visualization tool.
- [ ] Create a public or private project folder with a README.
- [ ] Schedule five 6-8 hour learning weeks.

| Spreadsheets  |  | Clean and summarize a messy dataset                  |
|---------------|--|------------------------------------------------------|
| SQL           |  | Answer business questions with joins and aggregation |
| Visualization |  | Create a focused dashboard or report                 |
| Statistics    |  | Explain variation and uncertainty responsibly        |
| Communication |  | Present a recommendation and limitations             |

## 02 - FOUNDATIONS

# Weeks 1-2: spreadsheet analysis

Use a public dataset with clear column definitions. Keep an untouched raw copy and document every transformation.

- ☐ Import and profile the data.
- ☐ Fix types, duplicates, missing values, and inconsistent categories.
- ☐ Use lookup, conditional, date, and aggregation functions.
- ☐ Create a pivot table that answers one stakeholder question.
- ☐ Build a one-page findings sheet with a caveat.

**Dataset, stakeholder question, and definition of done...**

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**Evidence**

Save the cleaned file, a data dictionary, before/after quality notes, and a one-page summary.

03 - QUERY

# Weeks 3-4: SQL

Practice on a relational dataset and treat query readability as part of the deliverable.

| Filtering        | WHERE, CASE, dates, and null handling               | [ ] |
|------------------|-----------------------------------------------------|-----|
| Aggregation      | GROUP BY, HAVING, rates, and distinct counts        | [ ] |
| Joins            | Explain row grain before and after each join        | [ ] |
| Window functions | Rank, running total, and period comparison          | [ ] |
| Validation       | Reconcile counts and investigate surprising results | [ ] |

Five questions my SQL project will answer...

Portfolio signal

Include commented queries and a short explanation of how you validated the result.

04 - REASON

# Weeks 5-6: statistics and metrics

Use descriptive statistics to understand the data before making recommendations. Do not imply causation from correlation.

- [ ] Define each metric and its grain.
- [ ] Compare mean and median where skew matters.
- [ ] Inspect distributions and outliers.
- [ ] Use rates with clear denominators.
- [ ] Segment carefully and check sample sizes.
- [ ] State at least two limitations.

## 05 - EXPLAIN

## Weeks 7-8: visualization

Every chart should help a named audience answer a question or make a decision.

- ☐ Use an accurate chart type.
- ☐ Write a takeaway title instead of a topic title.
- ☐ Label units, timeframe, and source.
- ☐ Use color to focus attention, not decorate.
- ☐ Check contrast and avoid relying on color alone.
- ☐ Include context, comparison, and caveats.

**Dashboard audience, decision, three key views, and one action...**

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06 - EXTEND

# Weeks 9-10: Python or automation

Add Python only if it appears in your target roles or helps make the workflow repeatable.

| Load     | Read data with a reproducible relative path          |
|----------|------------------------------------------------------|
| Clean    | Create functions for repeated transformations        |
| Validate | Assert expected columns, types, and ranges           |
| Analyze  | Produce a summary table used in the final report     |
| Export   | Write a clean dataset or figure without manual edits |

The manual step I will automate and why...

Skip safely

A strong SQL and spreadsheet project is better than a copied Python notebook you cannot explain.

07 - INTEGRATE

# Weeks 11-12: capstone

Build one small end-to-end analysis around a real decision. Keep the scope narrow enough to finish and revise.

| README                   | Question, audience, data source, approach, finding, limitation |
|--------------------------|----------------------------------------------------------------|
| Data dictionary          | Field definitions, units, grain, and important nulls           |
| SQL or notebook          | Readable, validated, and runnable from a clean start           |
| Dashboard or report      | Focused story with accessible labels                           |
| Five-minute presentation | Recommendation, evidence, caveat, next step                    |

Capstone question and intended audience...



08 - DAY 90

# Track progress and apply

Review evidence weekly and start conversations before the portfolio feels perfect.

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## Sources and responsible use

### U.S. Bureau of Labor Statistics - Occupational Outlook Handbook

Occupation duties, education, 2024 pay, and 2024-2034 outlook reference data.

<https://www.bls.gov/ooh/>

### O\*NET OnLine

Detailed occupation tasks, knowledge, skills, work activities, and technology examples.

<https://www.onetonline.org/>

### CareerOneStop

Career exploration, training, job-search, and local salary research tools.

<https://www.careeronestop.org/>

### BLS - Operations Research Analysts

A closely related analytical occupation: 2024 median pay and 2024-2034 outlook reference data.

<https://www.bls.gov/ooh/math/operations-research-analysts.htm>

#### Important

Career requirements, salaries, and hiring practices vary by country, city, employer, experience, and economic conditions. Use occupational data as a research starting point, not a promise of employment or pay.